



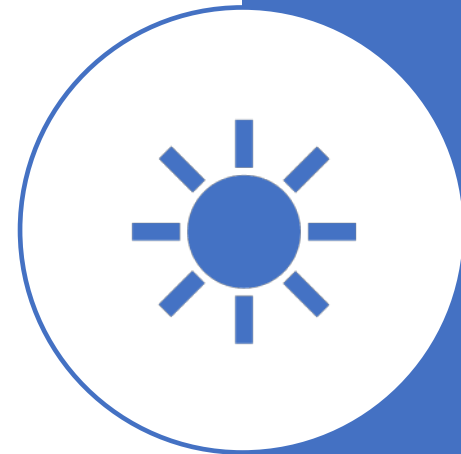
Solar Cells Materials

Dr. Eka Nurfani

PV as a Renewable Energy Source

Absorption of light, photon, creates an electric current, thus converting light directly into electric power, which is the most valuable form of energy.

- There are no moving parts, unlike conventional energy sources, which reduces the need for maintenance
- No fuel is necessary, this eliminates any environmental impact
- Long lifetime
- Modularity, The size of the plant can be increased depending on the requirements
- Decentralized power generation
- Minimized visual intrusion, can be integrated with the structures such as building



Photovoltaic solar energy

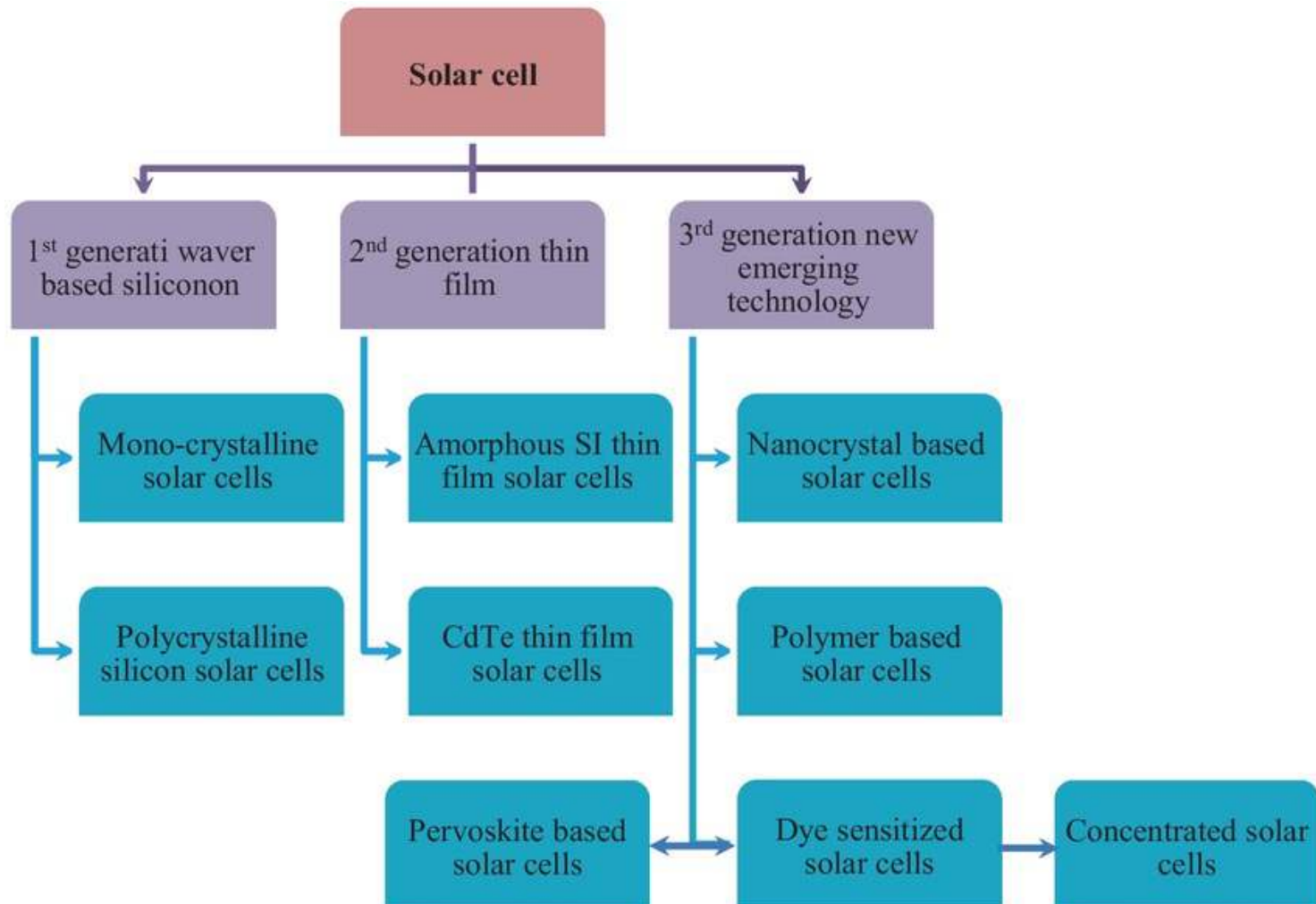
Advantages:

- environmentally friendly
- no noise, no moving parts
- no emissions
- no use of fuels and water
- minimal maintenance requirements
- long lifetime, up to 30 years
- electricity is generated wherever
- there is light, solar or artificial
- PV operates even in cloudy weather conditions
- modular “custom-made” energy can be sized for any application from watch to a multi-megawatt power plant

Limitations:

- PV cannot operate without light
- high initial costs that overshadow the low maintenance costs and lack of fuel costs
- large area needed for large-scale applications
- PV generates direct current or an inverter is needed
- an off-grid application energy storage is needed

Solar PV Technologies



Solar Cell Operation



The basic steps in the operation of a solar cell are: Generation of light-generated carriers;



Collection of the light-generated carries to generate a current;



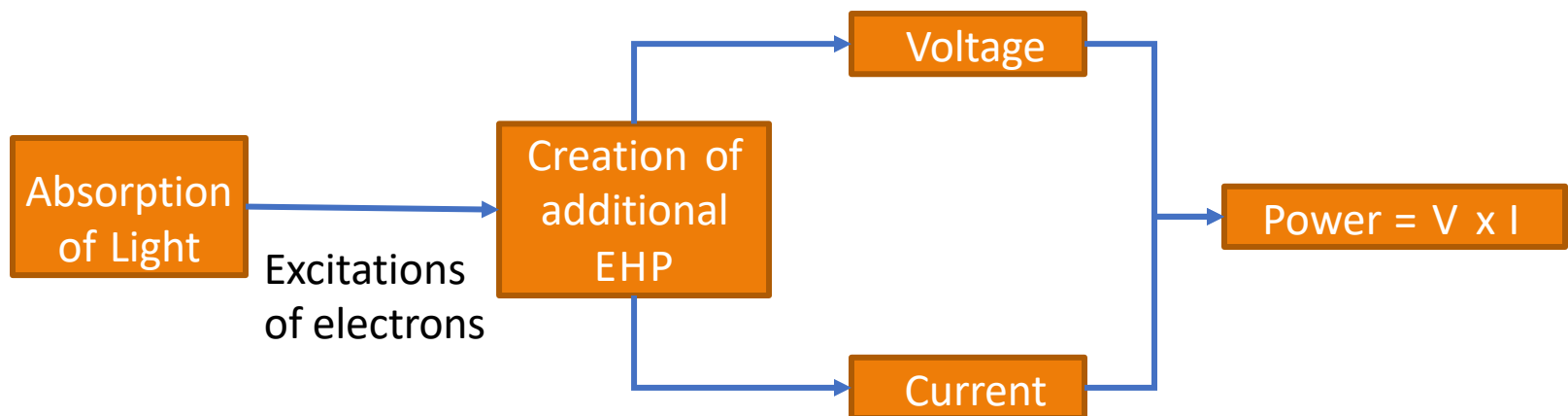
Generation of a voltage across the solar cell; and



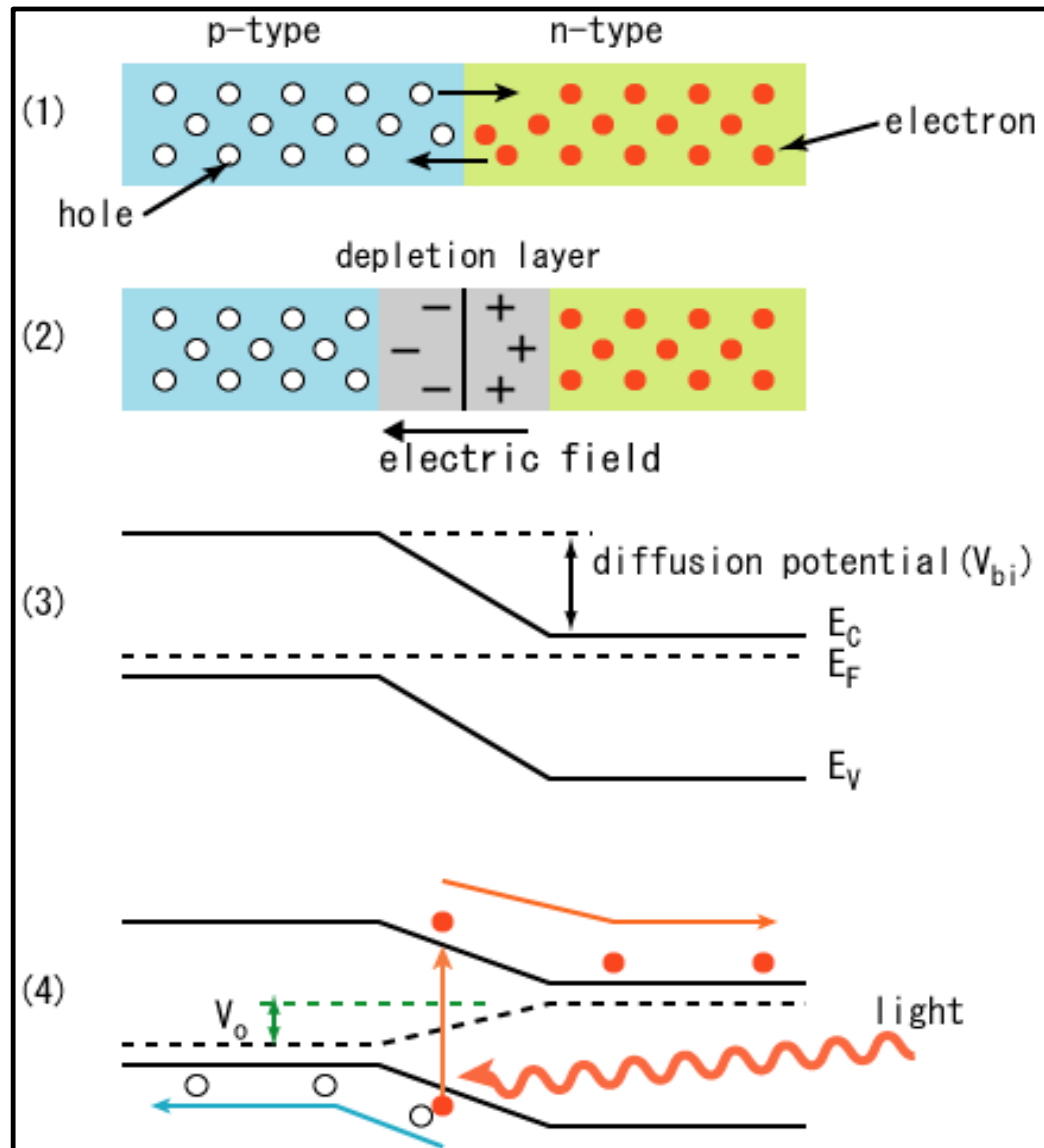
Dissipation of power in the load and in parasitic resistances.

Photovoltaic Effect

- Fundamental absorption is from:
 - absorption of photons: the excitation of an electron from the valence band to the conduction band
 - leaves a hole in the valence band
- Ideally, each incident photon with $E_{hv} > E_G$ will create one electron flowing in the external device.
- $E_{hv} < E_G$: semiconductor is transparent to light



Conventional p-n junction PV cells



Solar Cells

The photocurrent produces a voltage drop across the resistive load, which forward biases the pn junction.



Ideally, each incident photon with $E_{hv} > E_g$ will create one electron flowing in the external device.



$E_{hv} < E_g$: the device is transparent to the incident light.

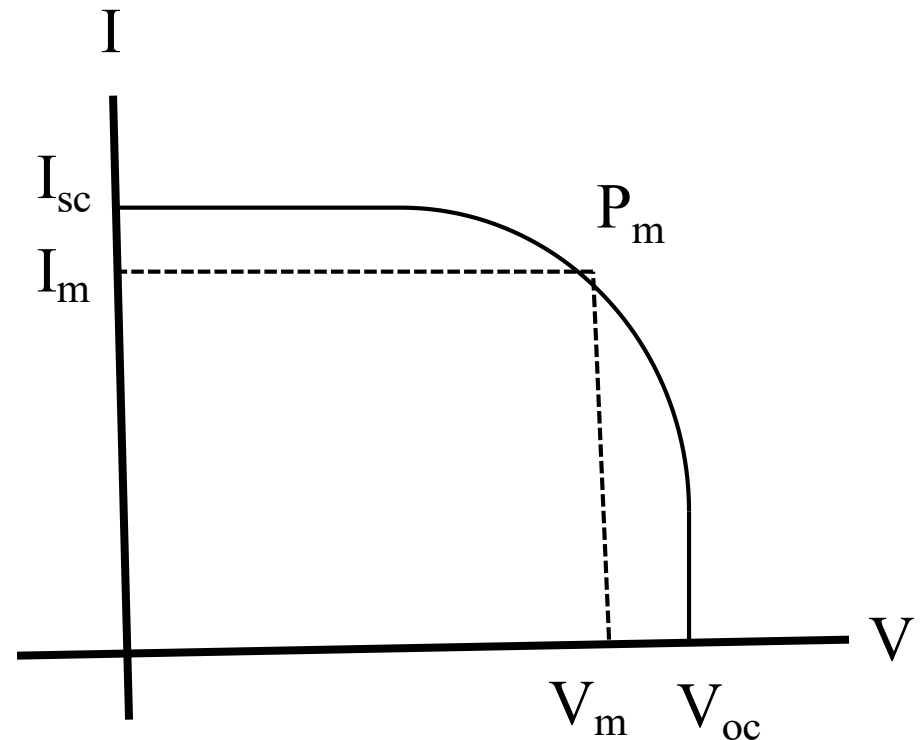
$E_{hv} \geq E_g$: photons are absorbed and EHP are photogenerated in the device.



$E_{hv} > E_g$: energy generated is lost as heat to the device.

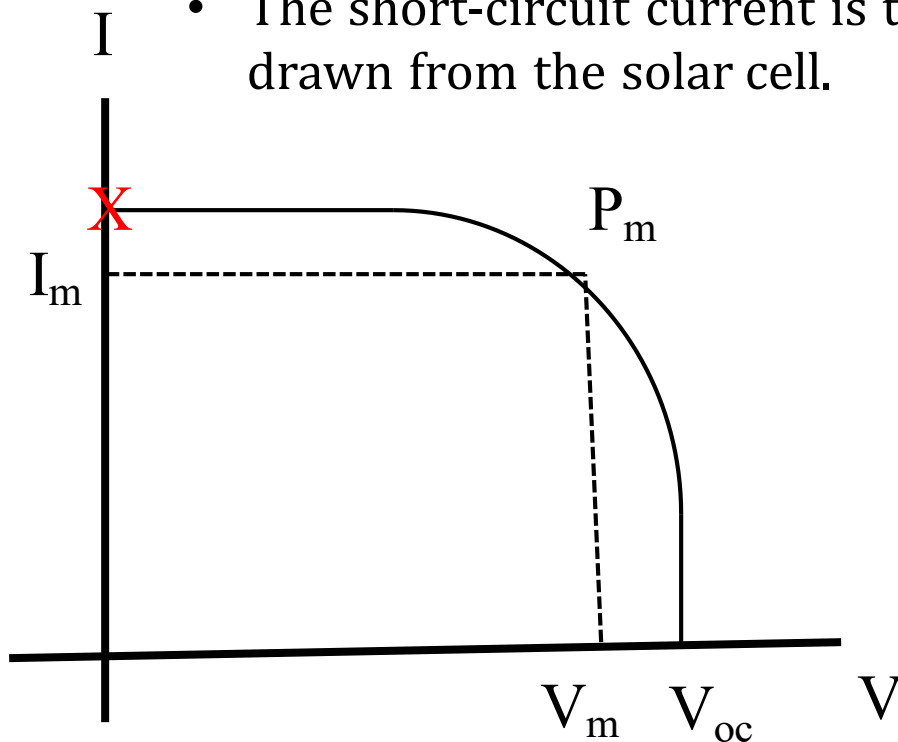
Solar Cell parameters

- V_{oc} —open circuit voltage
- I_{sc} —short circuit current
- P_m —maximum power point
- I_m , V_m —current and voltage at the maximum power point
- FF—Fill factor
- η —Efficiency
- R_s —series resistance
- R_{sh} —shunt resistance



Short-Circuit Current, I_{sc}

- The short-circuit current is the current through the solar cell when the voltage across it is zero (i.e., when the solar cell is short-circuited).
- The short-circuit current is due to the generation and collection of light-generated carriers.
- The short-circuit current is the largest current which may be drawn from the solar cell.



$$I_{sc} = q A W (L_p + L_n)$$

q - electronic charge

A - device area

W - depletion layer width

L_p - diffusion length in the p-type layer

L_n - diffusion length in the n type layer

Short-Circuit Current, I_{sc}

The short-circuit current depends on several factors :

- The area of the solar cell.
- The number of photons (i.e., the power of the incident light source).
- The spectrum of the incident light. For most solar cell measurements, the spectrum is standardized to the AM1.5 spectrum.
- The collection probability of the solar cell depends chiefly on the surface passivation and the minority carrier lifetime in the base.
- Optical properties (absorption and reflection) of the solar cell.

Short-circuit current dependence on The spectrum of the incident light

- The air mass represents the proportion of atmosphere that the light must pass through before striking the Earth relative to its overhead path length.
- Solar spectrum standards: AM 0, AM1, AM1.5

$$AM = \frac{1}{\cos\theta}$$

Where theta is the angle of sun rays

- AM 0 – solar radiation outside the earth atmosphere
- AM 1 - solar radiation when sun is exactly overhead
- AM 1.5 –when sun makes about 48 degree from zenith

Short-Circuit Current dependence on Absorption probability

- Blue light gets absorbed very close to the surface
- Green light penetrates more profound than the blue light
- Red light gets uniformly absorbed
- Penetration depth = $1 / \text{absorption coefficient}$
- A non-uniform generation causes a spectral dependence in the light-generated current

Short-circuit current and Quantum efficiency (QE)

1. QE is the ratio of the number of electrons collected by a solar cell per unit photon of a given energy (wavelength).
2. External quantum efficiency (EQE) and Internal quantum efficiency (IQE)
3. EQE is the probability of an incident photon contributing to one electron to the short circuit current

$$EQE(\lambda) = \frac{\Delta J}{q\Delta\Phi_{\lambda}}$$

Where J is current and Φ is the flux of photon

Short-circuit current and Quantum efficiency (QE)

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External quantum efficiency (EQE) and Internal quantum efficiency (IQE)

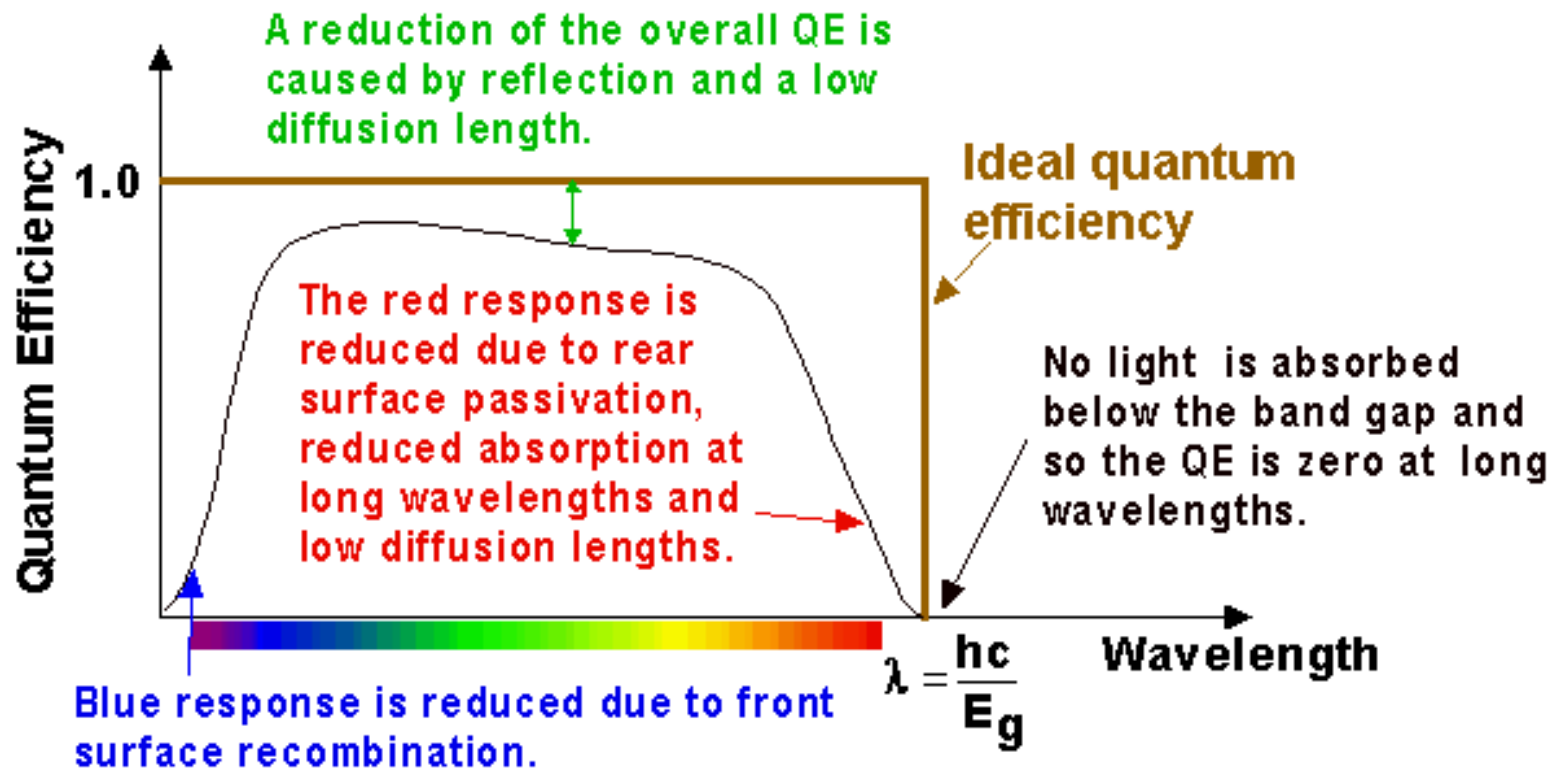
IQE is the probability of an absorbed photon contributing to one electron to the short circuit current

$$IQE(\lambda) = \frac{EQE}{1 - R(\lambda) - T(\lambda)}$$

Where R is reflectance and T is transmittance of the material

Short-circuit current and Quantum efficiency (QE)

QEs are function of wavelength and it depends on many parameters (absorption, reflection, surface properties, recombination)



Short-Circuit Current and Optical properties of surface

Photons in the spectrum can generate EHP, ideally, all the sunlight falling on the cell should be absorbed

Short circuit current is usually reduced due to optical losses

What are optical losses:

- Reflection
- Shadowing due to metal contact
- Partial absorption

Design criteria for small optical losses: minimize optical loss.

Open Circuit Voltage: Voc

The open-circuit voltage, V_{oc} , is the maximum voltage available from a solar cell, and this occurs at zero current.

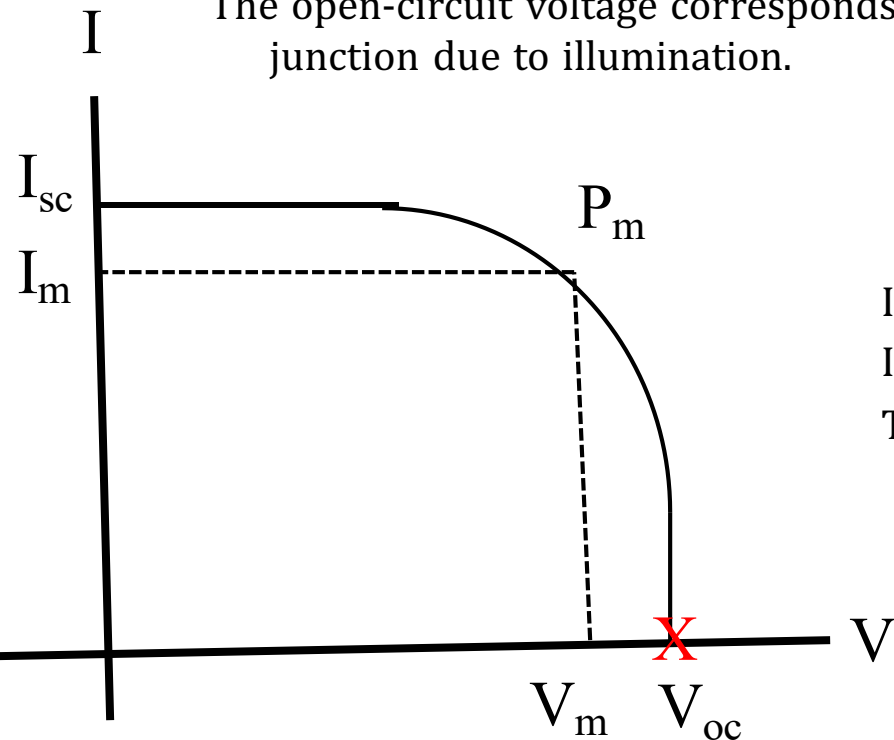
The open-circuit voltage corresponds to the amount of forward bias on the solar cell junction due to illumination.

$$V_{oc} = \frac{kT}{q} \ln \left(\frac{I_L}{I_0} + 1 \right)$$

I_L – light generated current, photocurrent

I_0 – reverse saturated current

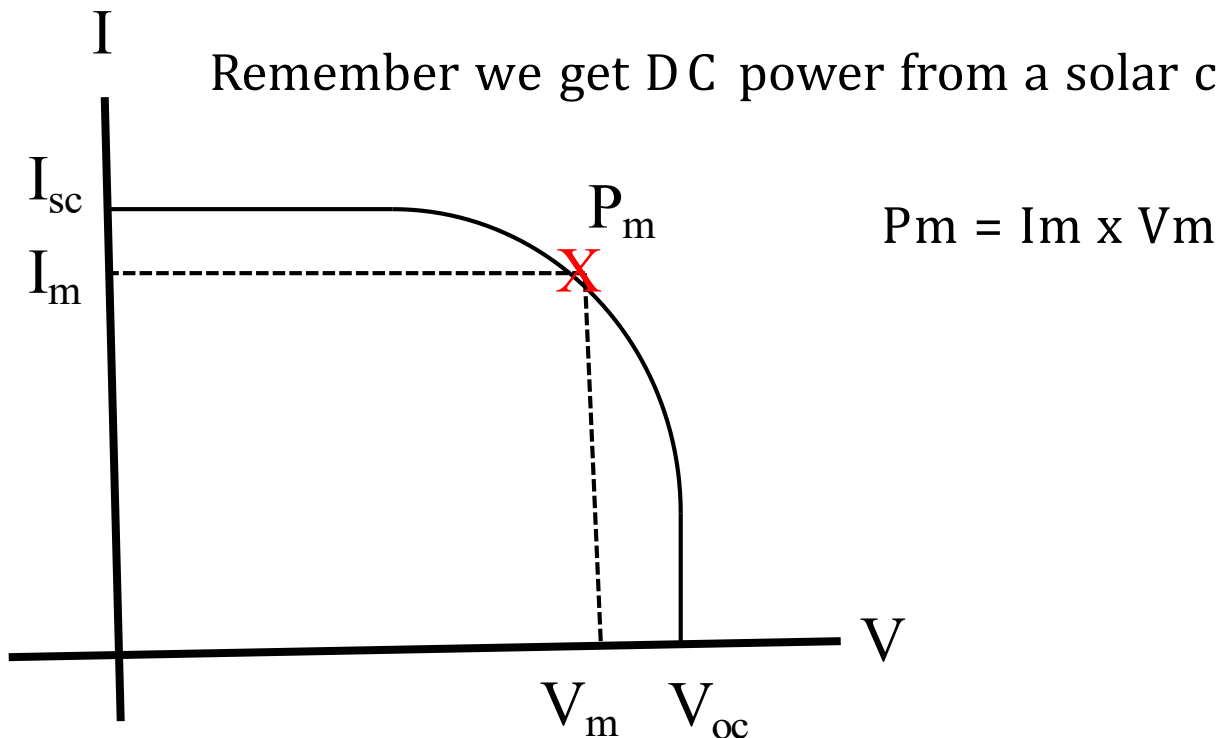
T – temperature k – Boltzmann constant



Maximum power: P_m

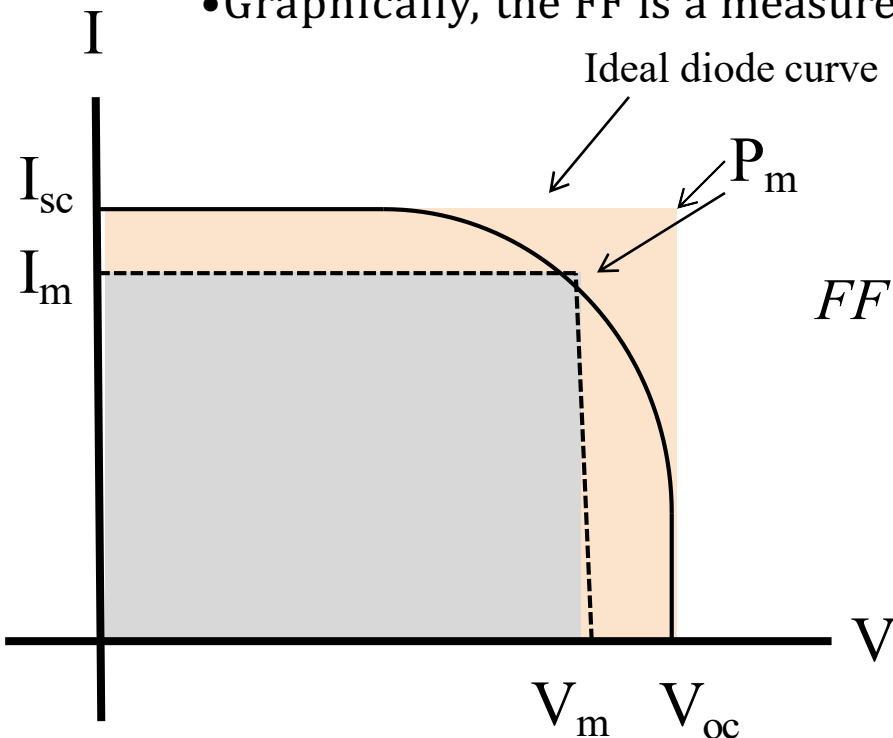
Power out of a solar cell increases with voltage, reaches a maximum (P_m) and then decreases again.

Remember we get DC power from a solar cell



Fill Factor: FF

- The FF is defined as the ratio of the maximum power from the actual solar cell to the maximum power from a ideal solar cell
- Graphically, the FF is a measure of the "squareness" of the solar cell.

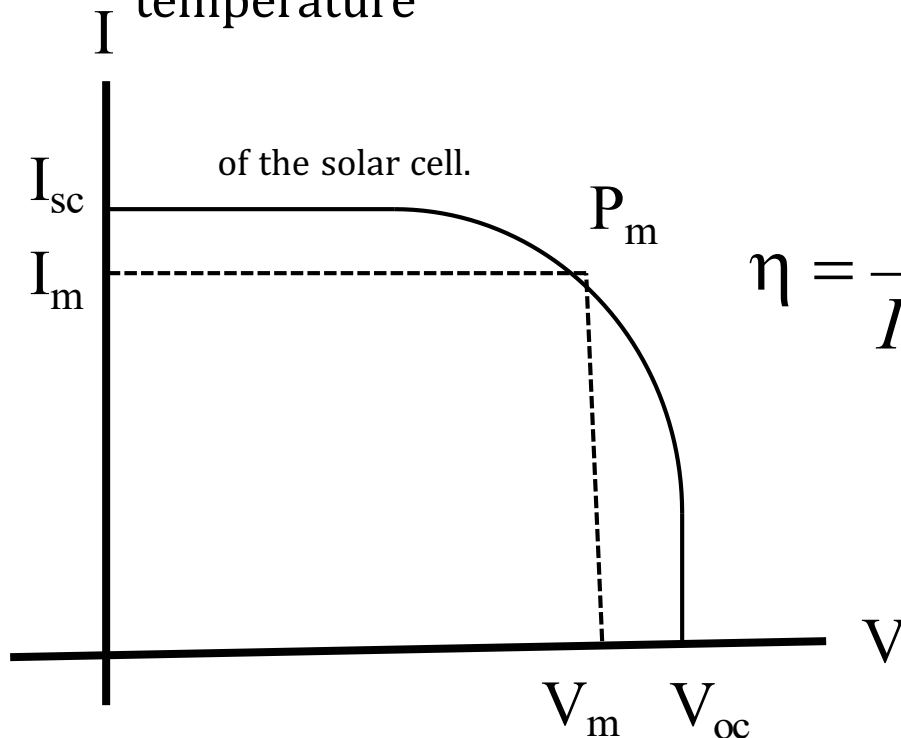


$$FF = \frac{\text{Max power from real cell}}{\text{Max power from ideal cell}} = \frac{V_m I_m}{V_{oc} I_{sc}}$$

Efficiency: η

Efficiency is defined as the ratio of energy output from the solar cell to input energy from the sun.

The efficiency is the most commonly used parameter to compare the performance of one solar cell to another. Efficiency of a cell also depends on the solar spectrum, intensity of sunlight and the temperature



$$\eta = \frac{\text{Max Cell Power}}{\text{Incident light Intensity}} = \frac{V_m I_m}{P_{in}}$$

$$\eta = \frac{V_{oc} I_{sc} FF}{P_{in}}$$



Enhancing the Performance of MAPbI_3 -Based Perovskite Solar Cells Fabricated Under Ambient Air: Effect of Cu, Ni, and Zn Doping into TiO_2

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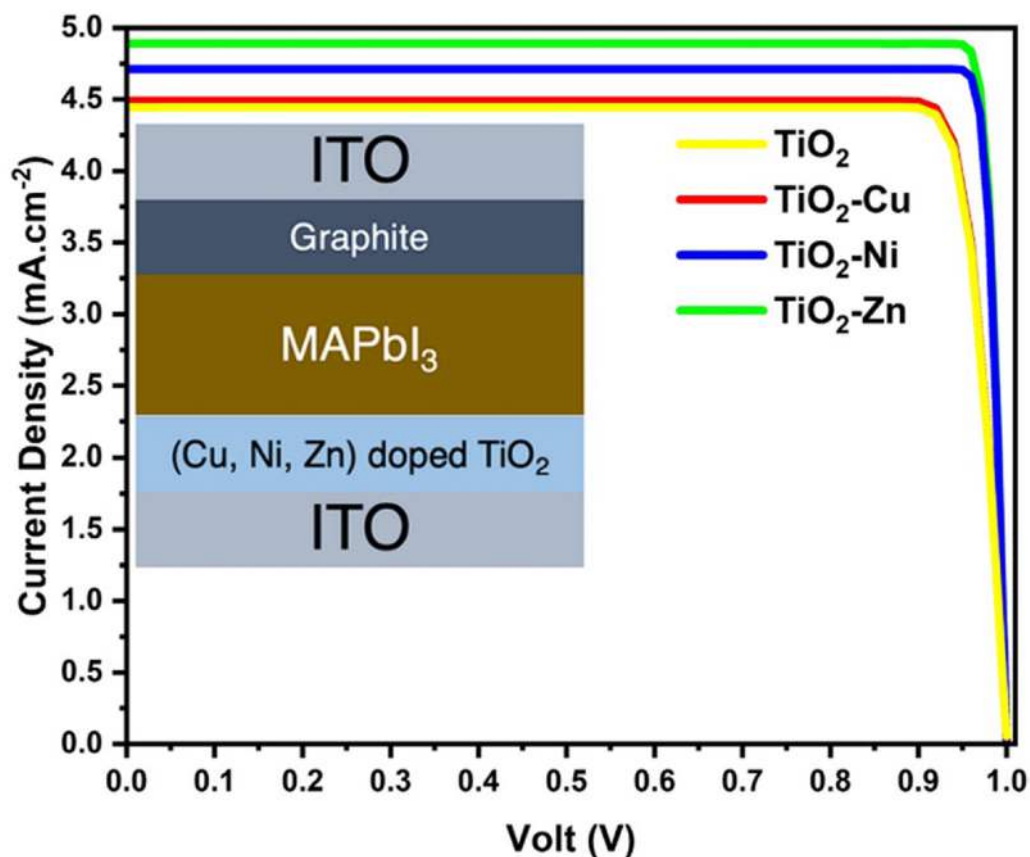


Fig. 6 J - V curves of fabricated MAPbI_3 -based PSC devices.



Research article

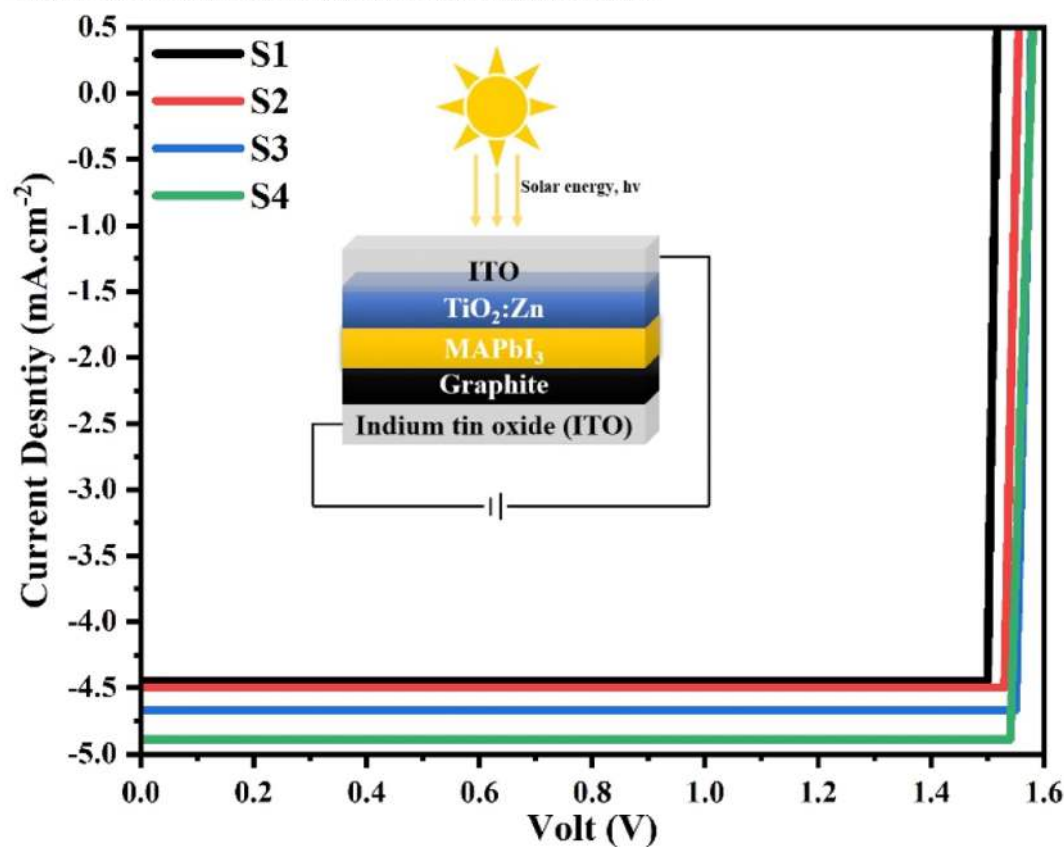
Effect of $\text{TiO}_2\text{:Zn}$ layer thickness on the performance of MAPbI_3 -based perovskite solar cells fabricated under open-air conditionMezan Adly Al Qadri^a, Eka Nurfani^{a,b,*}^a Department of Materials Engineering, Faculty of Industrial Technology, Institut Teknologi Sumatera (ITERA), Lampung 35365, Indonesia^b Center for Green and Sustainable Materials, Institut Teknologi Sumatera (ITERA), South Lampung 35365, Indonesia

Fig. 9. J-V curve of the fabricated devices.



PAPER

Effect of Mn doping on the performance of ZnO-based dye-sensitized solar cells

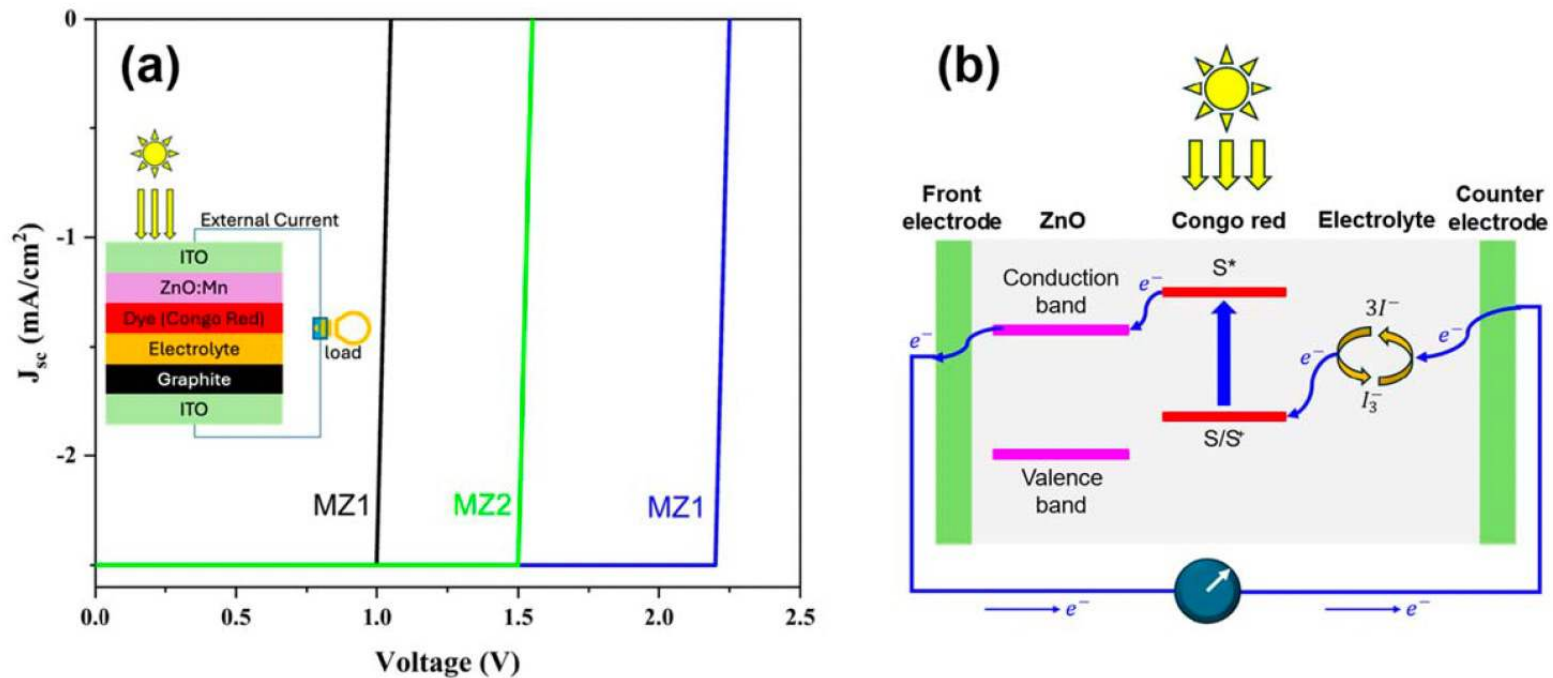
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30 April 2025Eka Nurfani^{1,2} , Viona R Putri¹, Mezan A Al Qadri¹, Wahyu S Sipahutar¹, Dedi Triyadi¹, Iwan S Saputra³,
Eri Widiyanto⁴ , Rishal Asri⁵ and Muhamad F Arif^{1,2}

Figure 5. (a) Current density versus voltage (J-V curves) for the MZ1, MZ2, and MZ3 samples. The inset of (a) is a schematic diagram of DSSC devices. (b) Working principle of DSSC.

Contoh Soal 1

Sebuah sel surya memiliki parameter sebagai berikut:

- **Short-Circuit Current (I_{sc}) = 3 A**
- **Open-Circuit Voltage (V_{oc}) = 0,6 V**
- **Fill Factor (FF) = 0,75**
- **Intensitas cahaya matahari yang diserap (P_{in}) = 1000 W/m²**
- **Luas sel surya (A) = 10 cm × 10 cm**

1. Hitung efisiensi sel surya tersebut!
2. Gambarkan kurva I-V dari ilustrasi di atas

Langkah 1: Hitung daya maksimum (P_{max})

$$P_{max} = V_{oc} \times I_{sc} \times FF$$

$$P_{max} = (0,6V) \times (3A) \times (0,75)$$

$$P_{max} = 1,35W$$

Langkah 2: Hitung daya input dari matahari (P_{in})

$$P_{in} = Intensitas \times Luas$$

$$P_{in} = (1000 W/m^2) \times (0,1m \times 0,1m)$$

$$P_{in} = 10W$$

Langkah 3: Hitung efisiensi sel surya

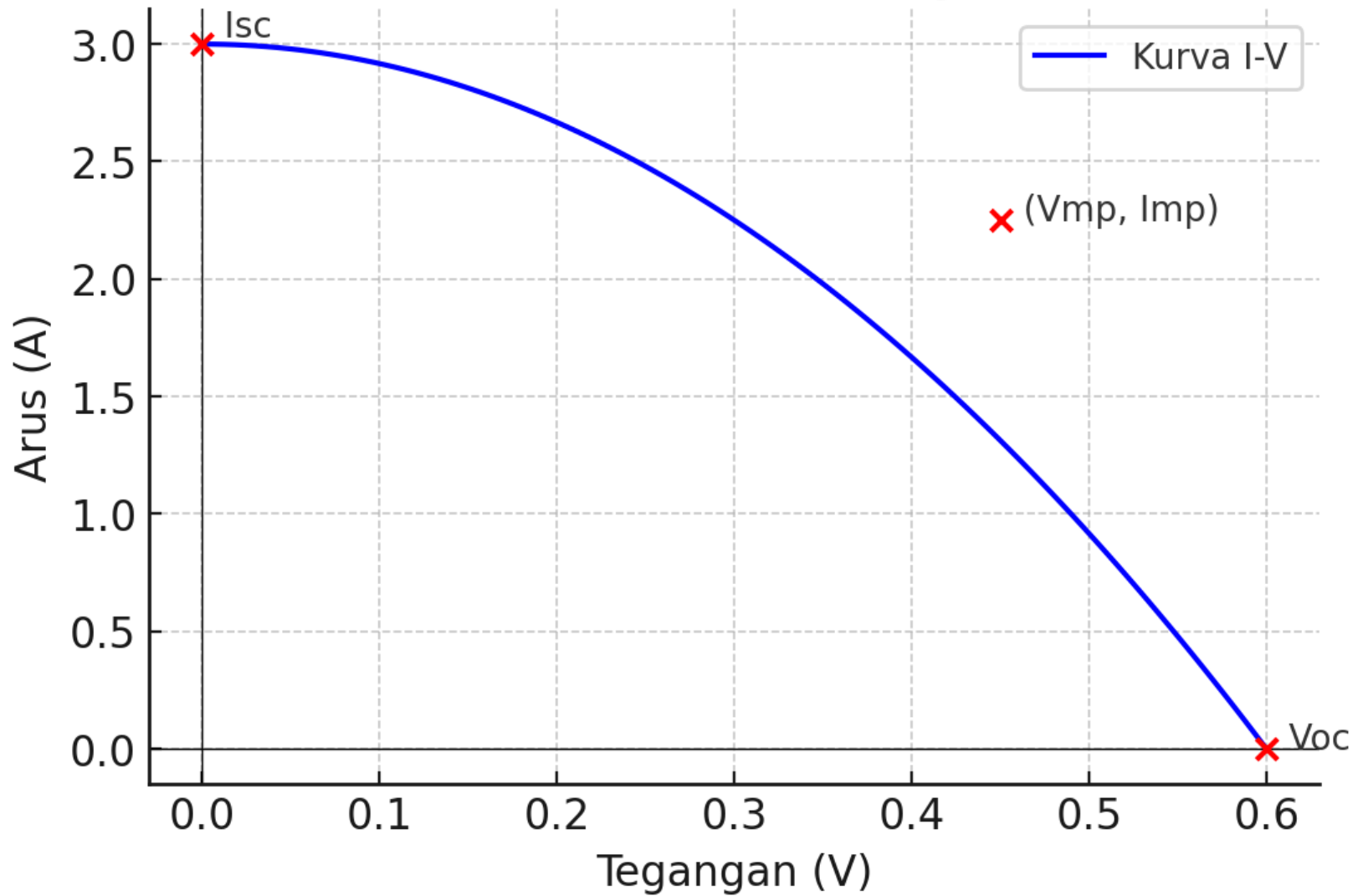
$$\eta = \frac{P_{max}}{P_{in}} \times 100\%$$

$$\eta = \frac{1,35W}{10W} \times 100\%$$

$$\eta = 13,5\%$$

Jadi, efisiensi sel surya tersebut adalah 13,5%. 

Kurva I-V Sel Surya



Efficiency is limited due to several factors.

- The energy of photons decreases at higher wavelengths.
- Radiation with higher wavelength causes only the heating up of solar cells and does not produce any electrical current.
- Each photon can cause only the production of one electron-hole pair.
- Self losses are caused by metal contacts on the upper side of a solar cell, solar cell resistance and due to solar radiation reflectance on the upper side (glass) of a solar cell.
- Other non-ideal parameters.

Non-Idealities

Bulk defects – dislocations and stacking faults, due to lattice mismatch with the substrate.

Surface recombination defects – EHP generated by the absorption of light can recombine before they cross the junction, therefore not contributing to the power output of the solar cell.

Bulk recombination defects – EHP generated further away from the junction have a large probability of recombining before they reach the device terminals.

Insufficient photon energy: $h\nu < E_g$

Excessive photon energy : $h\nu > E_g$

Solar cell is too thin – some of the light of the appropriate energy is not coupled into the cell and is passed through the device.

Open circuit Voltage (V_{oc}) losses – recombination of EHP in trap levels in the

depletion region that lowers V_{oc} .

Fill Factor losses – related to V_{oc} , series resistance, and shunt resistance.

Reflection losses.

Minimizing optical losses

Top contact coverage of the cell surface can be minimized.

Anti-reflection coatings can be used on the top surface of the cell. Reflection can be reduced by surface texturing.

The solar cell can be made thicker to increase absorption.

The optical path length in the solar cell may be increased by a combination of surface texturing and light trapping.

Anti-Reflection Coating

- Prevents incident light from reflecting off of the device.
- The AR coating needs to have the correct refractive index for the material system and be transparent.
- Deposited as noncrystalline or amorphous layer which prevents problems with light scattering at grain boundaries.
- A double layer AR coating reduces the reflection of usable sunlight to $\sim 4\%$.

Choice of ARC

The thickness of an ARC is chosen such that the reflected wave has destructive interference, resulting in zero reflected energy.

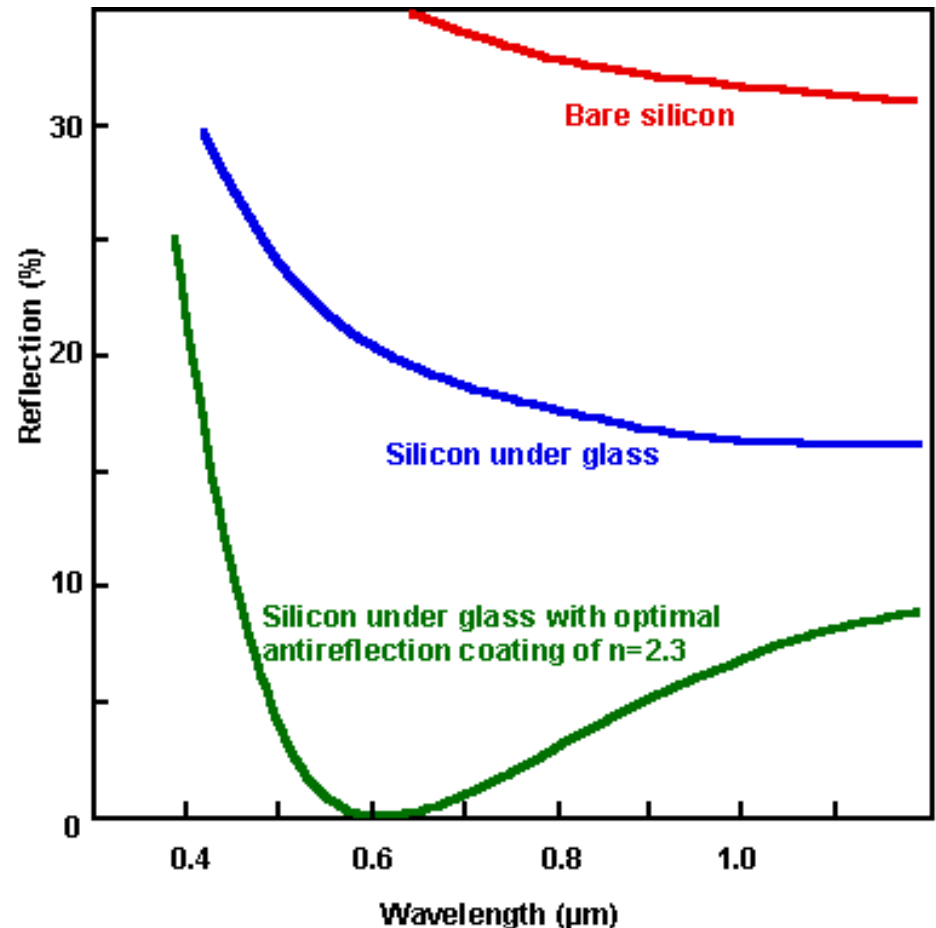
The thickness of the ARC is chosen so that the wavelength in the dielectric material is one-quarter the wavelength of the incoming wave (destructive interference).

Reflection from various combination

Index of refraction is also a function of wavelength, minimum reflection is obtained for one wavelength.

Multilayer structure reduces the reflection losses.

More than one ARC can be used, but expensive.

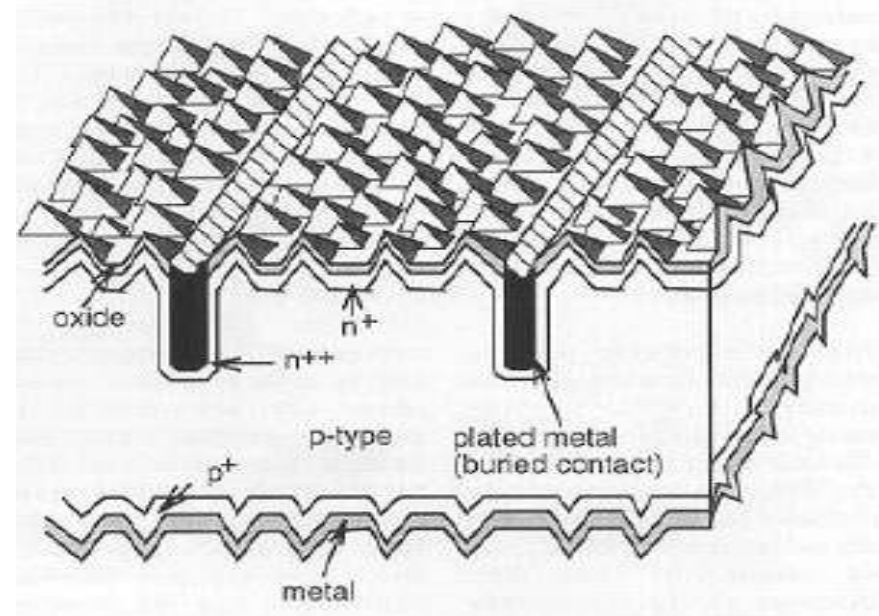
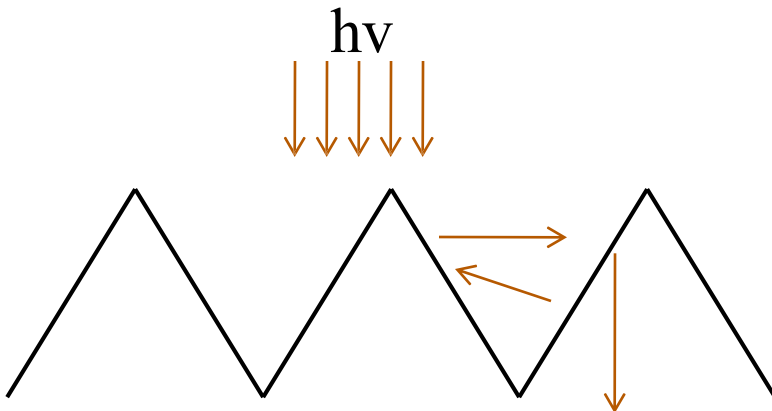


Si Technology

Textured top layer

Incident light will:

- Become trapped
- Bounced around in the texture
- Absorbed in the device



Surface texturing

Any rough surface decreases the reflection by increasing the chances of the reflected rays bouncing back on the surface

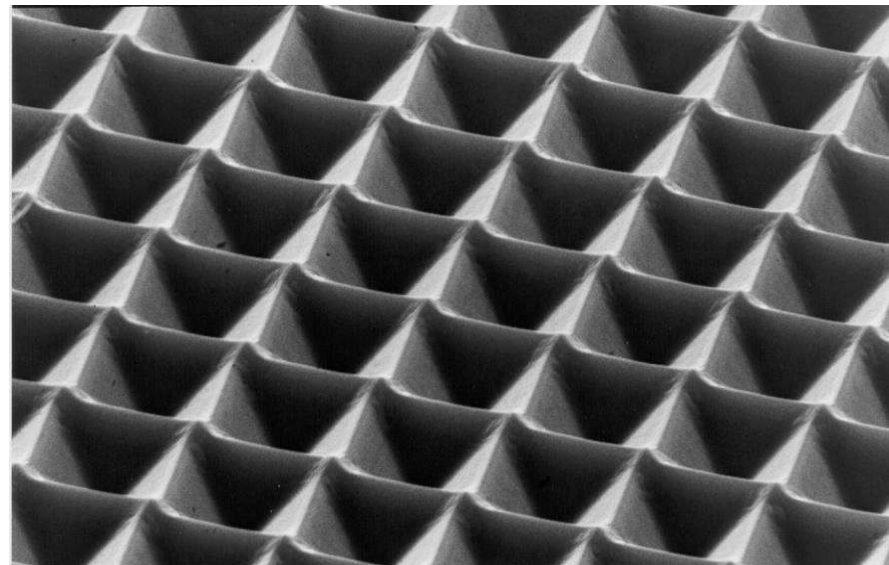
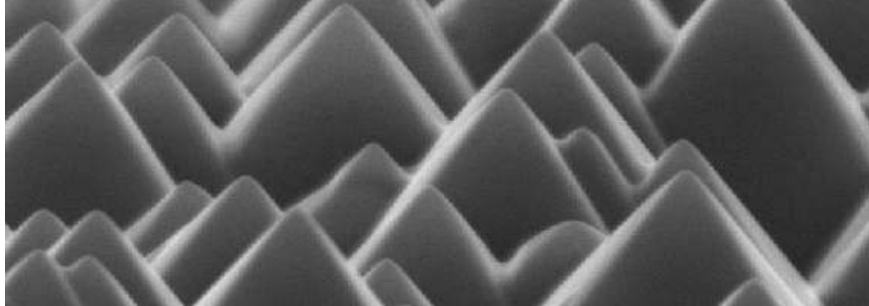
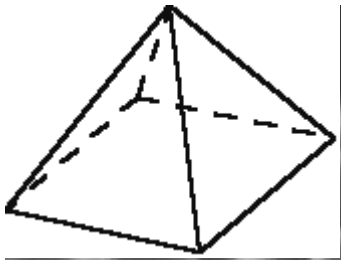
Surface texturing can be obtained by selective etching → a process by which material is removed by chemical reaction

Selective etching is based on the concept of different orientation properties in different direction in crystals

Etching rate are different in $\langle 100 \rangle$ dir than in $\langle 111 \rangle$ dir

Surface texturing

- Chemical etching in KOH results in pyramid formation on the Si surface → etching is faster in $\langle 100 \rangle$ direction than in $\langle 111 \rangle$ direction
- Using photolithography, inverted pyramids can be obtained, which are even more effective



Light trapping

- Rear side reflector or rear side texturing is used to increase the optical path length in solar cell
- Increased optical path is required for thin solar cell (thin solar cell have higher Voc. It saves expensive Si)
- Total internal reflection (TIR) condition are used to increase the optical path length

Current loss due to recombination

Recombination of carriers reduces both short-circuit current as well as open-circuit voltage.

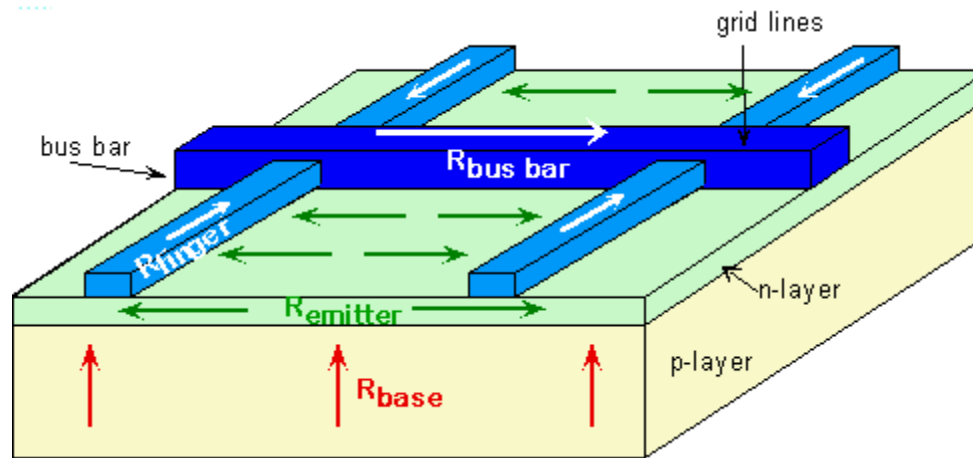
Recombination areas

- Surface recombination
- Bulk recombination
- Depletion region recombination

Design criteria: The carrier must be generated within a diffusion length of the junction so that it will be able to diffuse to the junction before recombining.

Resistive Losses

Resistive effects (series and shunt resistance) in solar cells reduce the efficiency of the solar cell by dissipating power in the resistances. Both the magnitude and impact of series and shunt resistance depend on the geometry of the solar cell and solar cell area



Contributing factors to R_s :

1. the movement of current through the emitter and base of the solar cell
2. the contact resistance between the metal contact and the silicon
3. resistance of the top and rear metal contacts

Solar Cells

Resistance

There are electrodes attached to the n-side which allow illumination and form an array of electrodes.

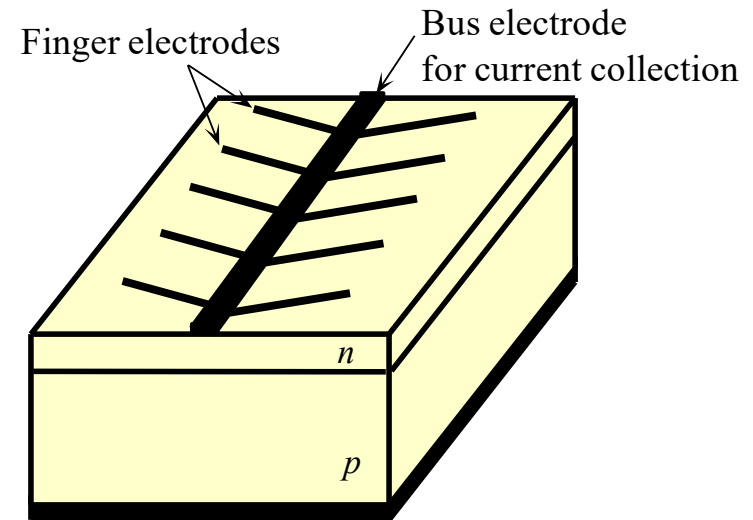
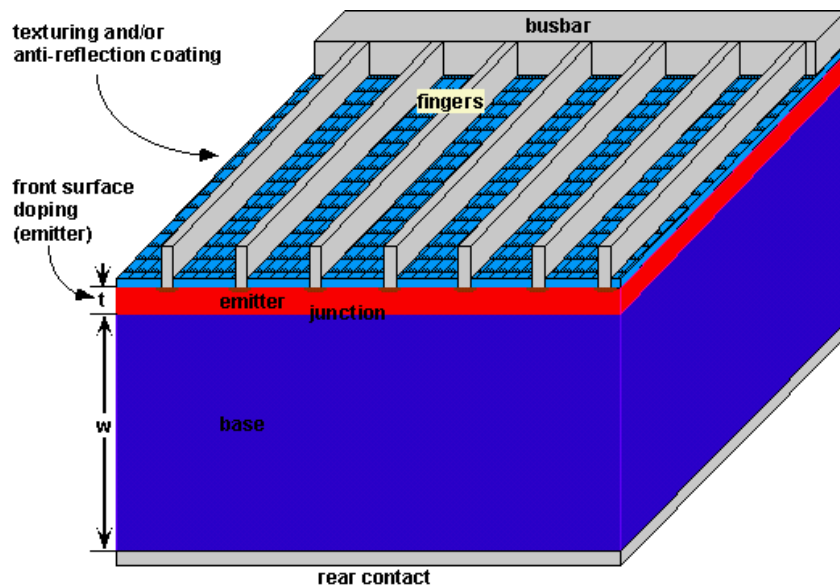
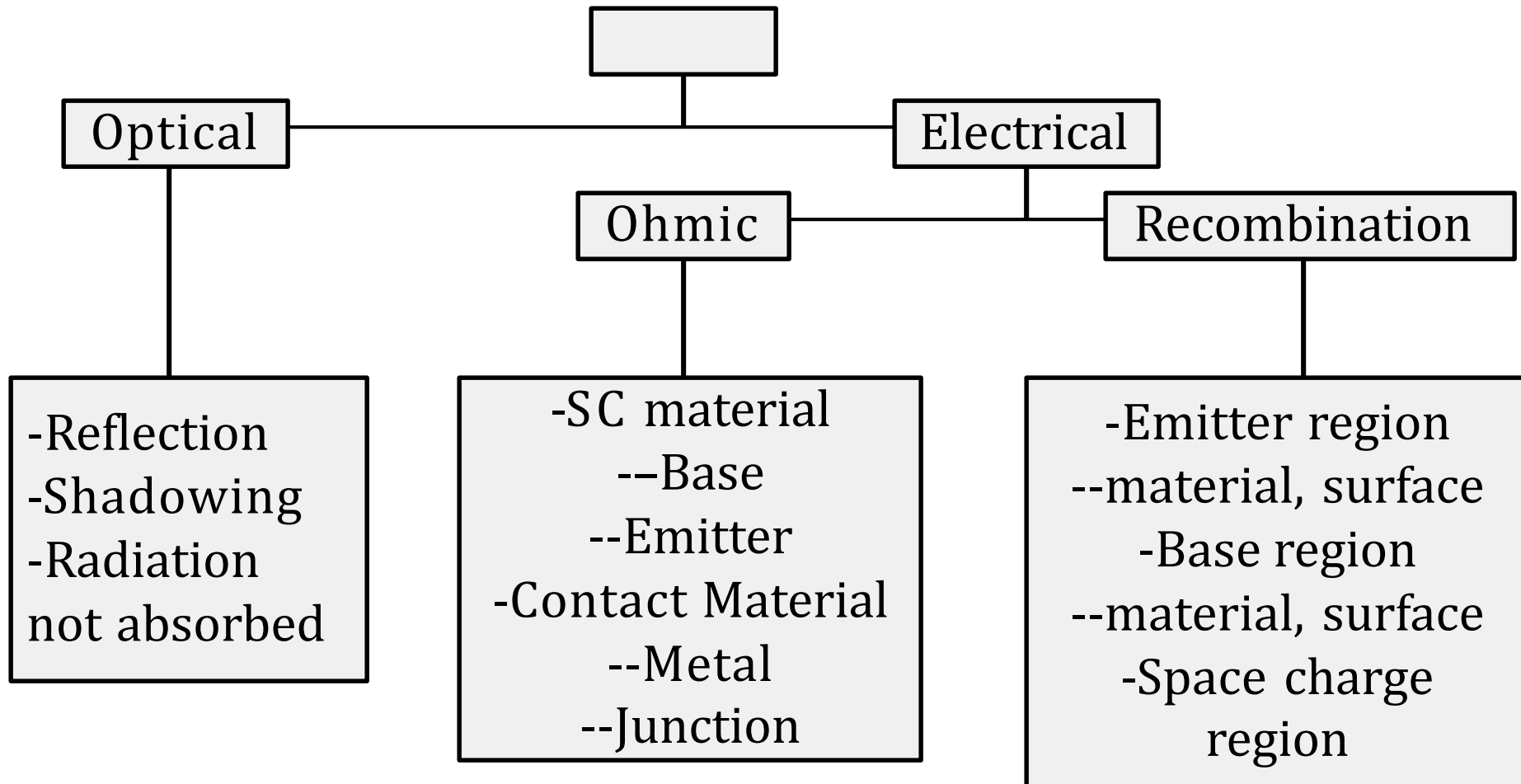


Fig. 6.50: Finger electrodes on the surface of a solar cell reduce the series resistance

Summary of losses in Solar cell Loss



Challenge

To provide the world with 10 TW of solar electricity by 2030

Need to grow the industry by $\sim 35\%$ /year.

so far it has

Not run out of essential materials.

problem?

Make enough money in 2 years to double the factory size.

profits must be made

Get energy payback within two years so that we generate more power than we use.

1 year payback is not here yet

Requirement for ideal solar cell material

Bandgap between 1.1
to 1.7 eV

Direct band structure

Consisting of readily
available, non-toxic
material

Easily reproducible
deposition techniques,
suitable for large-area
production

Good PV conversion
efficiency

Long-term stability